Synchronous TMR Architecture with Single-Cycle FDI for Radiation-Prone FPGA DAQ Systems

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Repo: github.com/faizakhosol-prog/Resilient-FPGA-Muon-DAQ

Problem

Radiation-induced Single Event Upsets (SEUs) flip FPGA state bits mid-operation, corrupting critical control paths and sequential logic in unshielded, safety-critical DAQ pipelines like cosmic-ray muon detection.

Solution

Independently engineered a fully synchronous Triple Modular Redundancy (TMR) control core in VHDL on an AMD Xilinx Artix-7 FPGA (xc7a100tcsg324-1, Nexys 4 DDR). Three parallel hardware core replicas are masked by a glitch-free registered majority voter that filters transient routing hazards. A parallel, compact Fault Detection & Isolation (FDI) telemetry engine flags discrepancies and isolates the corrupted core index (01=Core A, 10=Core B, 11=Core C) within exactly one clock cycle, providing a clean interface for external scrubbing or Dynamic Function eXchange (DFX) controllers.

Key Results

- **Timing Closure:** Successfully met all constraints at 100 MHz post-implementation (Worst Pulse Width Slack, WPWS = 4.500 ns, with zero failing endpoints).
- **SEU Simulation Validation:** Verified via Vivado behavioral simulation with multi-cycle runtime fault injection. When Core A was corrupted at 250 ns, the voter successfully held the correct state; the fault was flagged at the next clock edge, and the system continued uninterrupted. Core A state vectors were successfully realigned via external testbench recovery emulation at 450 ns with zero data loss and zero system downtime.
- **Ultra-Low Footprint:** Consumes an exceptionally minimal footprint of only 7 LUTs and 5 Flip-Flops (under 0.01% of the Artix-7 fabric).
- **Power Efficiency:** Core internal logic power footprint is just 33 mW, ensuring excellent thermal overhead for remote edge deployments.

Integration & Impact

The architecture is designed to scale linearly to protect critical control paths (projecting to ~48 FFs and 52 LUTs for a 16-bit TDC counter layer, keeping total overhead under 0.1% of the fabric). By keeping fault tracking within the synchronous boundary, the design avoids heavy frequency degradation while generating cycle-accurate status flags, making it a deterministic front-end for active, fault-aware hardware control systems. *Manuscript in preparation.*